Notes from Ben, not part of the talk

- Side chain. Explain backbone and side chain once, on slide 2 or 4, before they are named (Fig. 1c, Fig. 2b, slide 10).

Highly accurate protein structure prediction with AlphaFold

Paper by Jumper et al., *Nature* 596 (2021)

Evita Bosch Pokunevich Ben Bullinger

Machine Learning for Genomics (263-5351-00L), ETH Zürich, Fall 2026

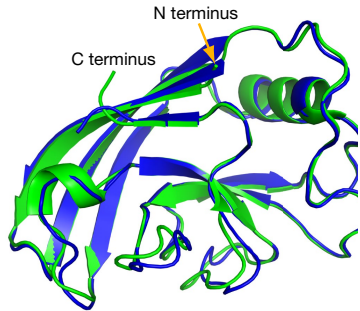
Outline

1. The problem
2. The benchmark
3. The network
   - The data
   - The network, in detail
4. Training
5. What makes it work
6. Conclusion
7. Questions

Sequence in, structure out

SIFSYITESTGTPS…

sequence of amino acids



3D coordinates of all heavy atoms

Predicted (blue) vs experimental (green) structures

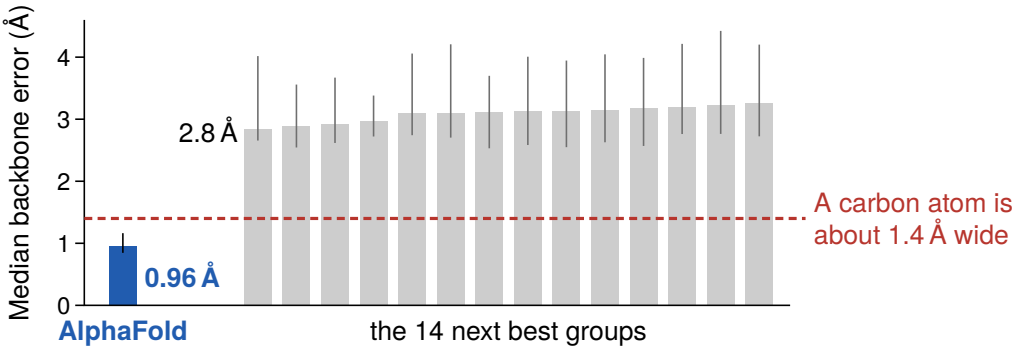
An experiment takes months to years

Billions of sequences are known

The problem is more than 50 years old

Sequence and structure are T1049. The structure is Fig.1b (main paper)

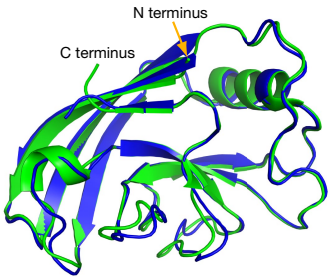
CASP14: atomic accuracy, alone



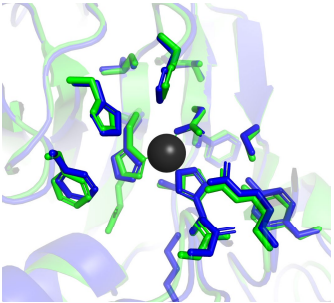
The median backbone error is below one atom width

Redrawn from Fig. 1a. The error is the Cα r.m.s.d.$_{95}$ on 87 domains, with 95% confidence intervals.

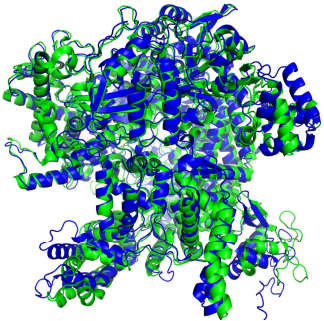
Predictions on top of experiment



Backbone



Side chains

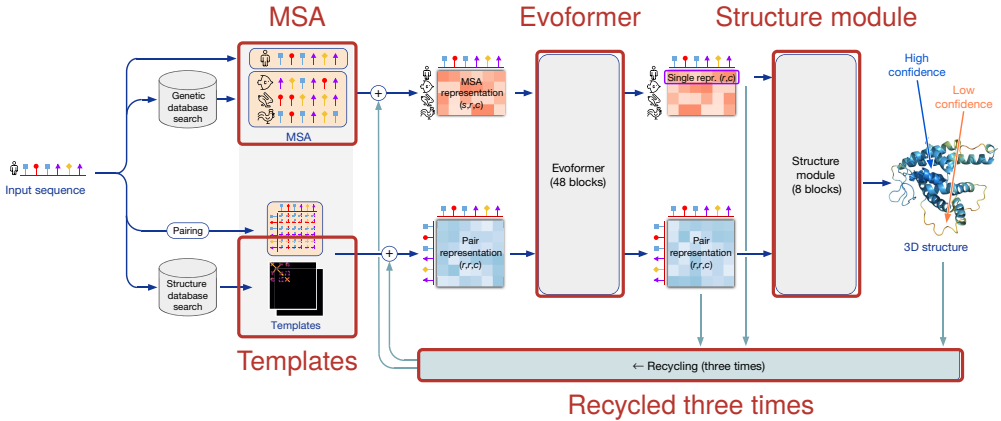


A 2,180-residue chain

AlphaFold experiment

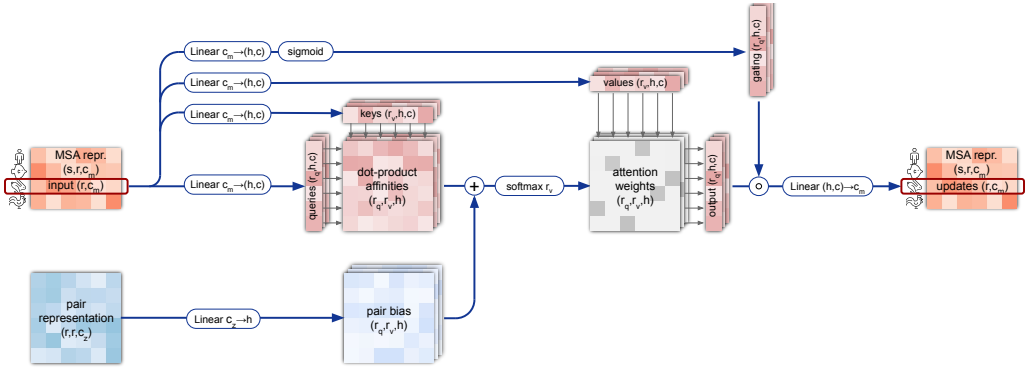
Prediction and experiment overlap, also in the side chains

One pipeline, end to end



Coordinates come out of the network itself

One pipeline, end to end

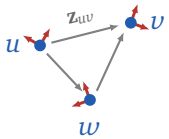


Coordinates come out of the network itself

AlphaFold 2 is message passing on a geometric graph

$$\mathbf{h}_u = \phi\Big(\mathbf{x}_u, \bigoplus_{v \in \mathcal{N}_u} \psi(\mathbf{x}_u, \mathbf{x}_v, \mathbf{e}_{uv})\Big)$$

message ψ, aggregation $\bigoplus$, update ϕ (Bronstein et al. 2021).
Nodes: residues, $\mathbf{s}_u$ and frame T_u. Edges: $\mathbf{z}_{uv}$.

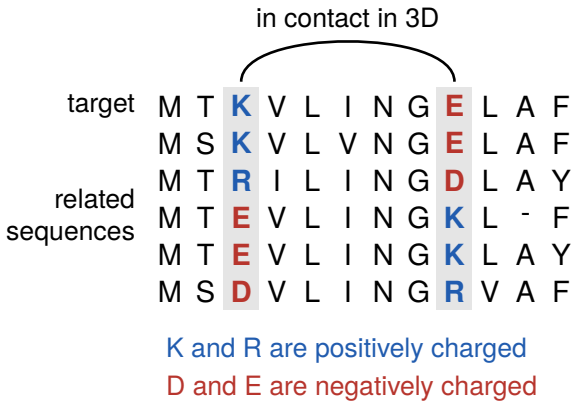


Block	message ψ	into
MSA row attention	$\psi(\mathbf{m}_{su}, \mathbf{m}_{sv}, \mathbf{z}_{uv})$, the edge biases the weight	node u
outer product mean	$\psi(\mathbf{m}_{\cdot u}, \mathbf{m}_{\cdot v})$, averaged over the sequences	edge uv
triangle updates	$\psi(\mathbf{z}_{uw}, \mathbf{z}_{wv})$ over every third node w	edge uv
IPA	$\psi(\mathbf{s}_u, \mathbf{s}_v, \mathbf{z}_{uv}, \lVert T_u \circ \vec{\mathbf{q}}_u - T_v \circ \vec{\mathbf{k}}_v \rVert)$, values moved by $T_u^{-1} \circ T_v$	node u
backbone update	none, ϕ alone	frame T_u

Triangles are two-hop messages. They make the edges consistent with one structure.

Geometry enters only through distances. A frame is a gauge fixed by its atoms.

Input: a multiple sequence alignment (MSA)



A row is a related protein

A column is an aligned position

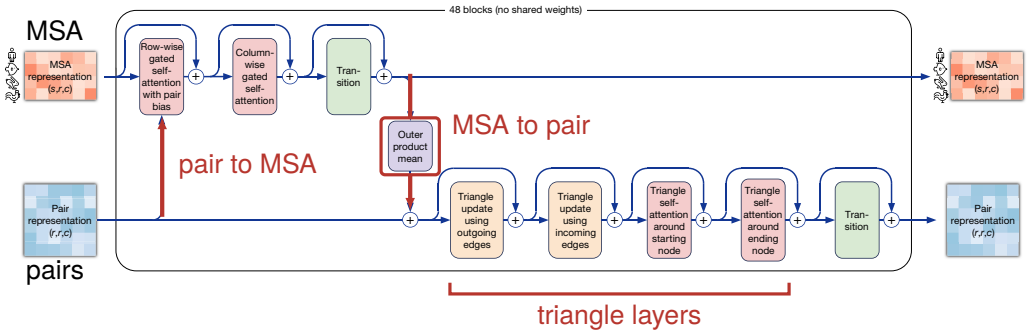
Contacts may change together

The network reads the sequences themselves

D. S. Marks et al., *PLoS ONE* 6, 2011. The alignment is an illustration, not data.

Evoformer block



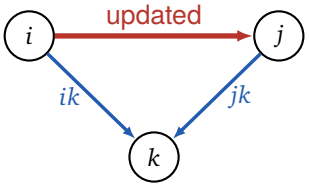


The 48 blocks do not share weights

MSA and pair representation exchange information

Pairs updated through triangles



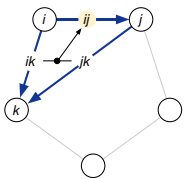


Edge ij is updated from the edges ik and jk

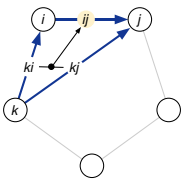
The triangle inequality acts as an inductive bias

Multiplication

outgoing edges

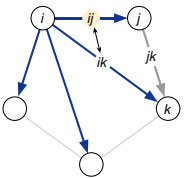


incoming edges

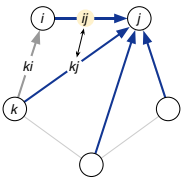


Attention

start node

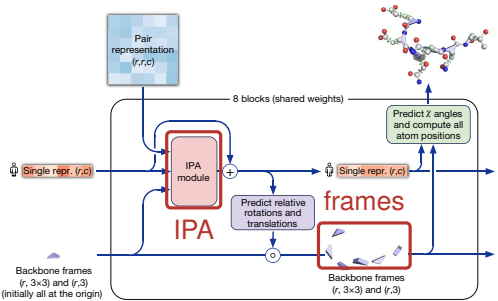


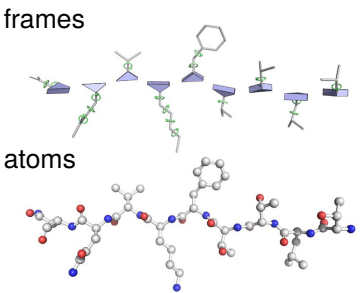
end node



Structure module: residues as rigid bodies







A frame is one rotation and one translation per residue

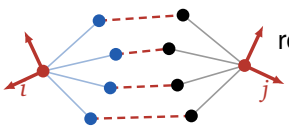
All frames are updated at once

Nothing forces neighbouring residues to stay connected

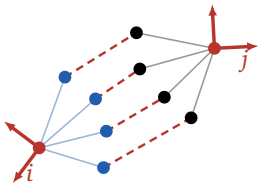
Invariant point attention



$$a_{ij} = \text{softmax}_j\Big(\underbrace{\boldsymbol{q}_i^\top \boldsymbol{k}_j}_{\text{usual attention}} + \underbrace{b_{ij}}_{\text{pair bias}} - \underbrace{\textstyle\sum_p \left\| T_i \circ \vec{\boldsymbol{q}}_{ip} - T_j \circ \vec{\boldsymbol{k}}_{jp} \right\|^2}_{\text{distances between 3D points}} \Big)$$



rotate everything



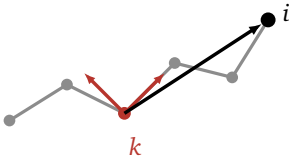
Points are placed in the frames of i and j

The distances stay the same

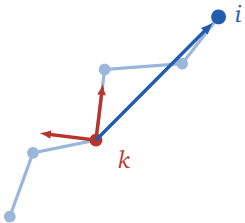
The weights are invariant to global rotation and translation

The formula is Algorithm 22 of the supplement for one head, without scaling constants.

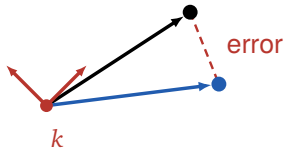
FAPE: a loss over local frames



experiment



prediction



both, seen from frame k

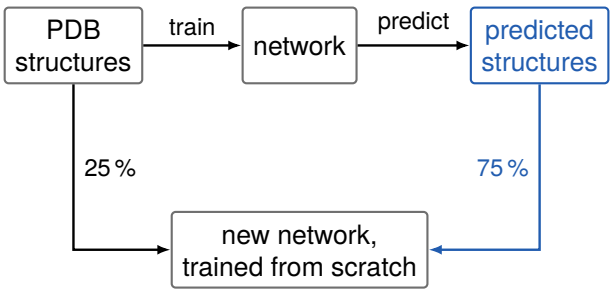
$$\text{FAPE} = \frac{1}{Z} \underset{k,i}{\text{mean}} \min\Big(d_{\text{clamp}},\ \big\|(\boldsymbol{R}_k, \boldsymbol{t}_k)^{-1}\boldsymbol{x}_i - (\boldsymbol{R}_k^{\text{true}}, \boldsymbol{t}_k^{\text{true}})^{-1}\boldsymbol{x}_i^{\text{true}}\big\|\Big)$$

A rotation changes nothing. A mirror image is penalised.

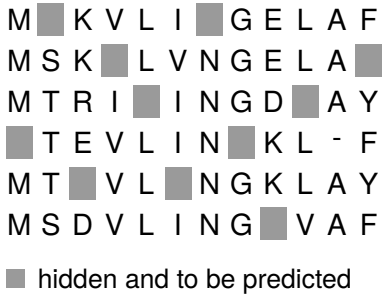
Supplementary Information, Algorithm 28, with $Z = d_{\text{clamp}} = 10\,\text{Å}$. The drawing is an illustration.

Trained beyond the PDB

Self-distillation adds 350,000 structures



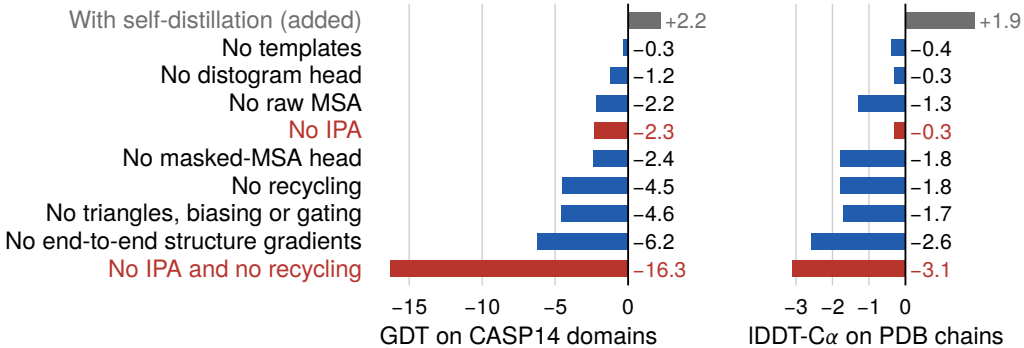
A masked-MSA loss is added



Training uses sequences without structures

Self-distillation follows Q. Xie et al., 2020. Masking follows BERT, J. Devlin et al., 2019.

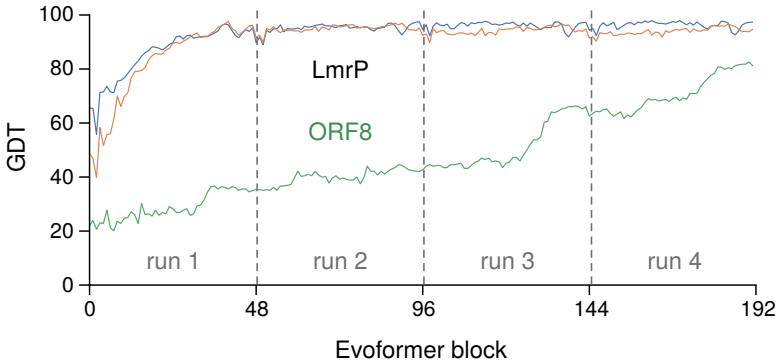
Which pieces earn their accuracy



Many components contribute

Redrawn from Fig. 4a. A bar is the mean change against the baseline over the training runs of a row.

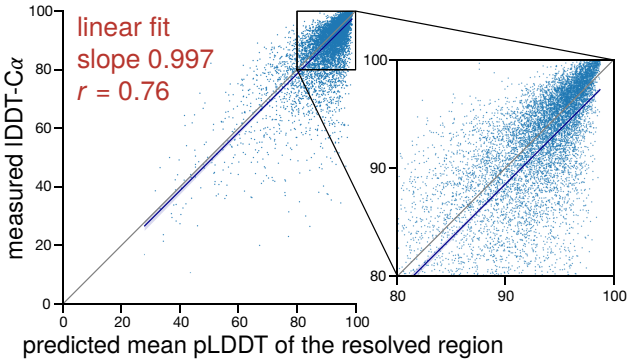
The structure forms inside the trunk



The trunk is frozen and every block gets a structure module

Some proteins are solved early and others late

The model knows when it is right

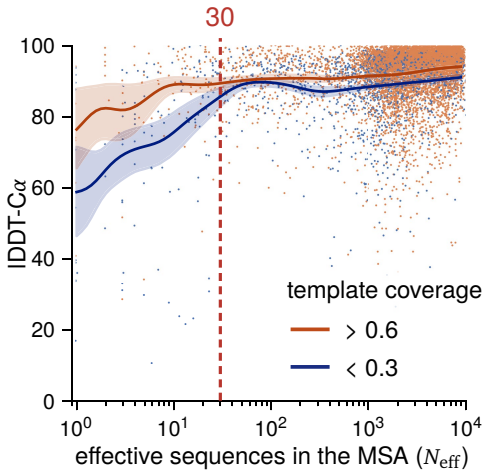


pLDDT is the predicted lDDT-Cα per residue

Predictions need a well-calibrated confidence

K. Tunyasuvunakool et al., *Nature* 596, 2021.

Where it still fails



Accuracy falls below about 30 effective sequences

AlphaFold is weaker for chains shaped by other chains

AlphaFold needs enough related sequences

What followed

2021 — RoseTTAFold appears online the same day

98.5% of human proteins are predicted

AlphaFold-Multimer predicts complexes

2022 — ESMFold uses a language model and no MSA

2024 — AlphaFold 3 is one model for proteins, DNA, RNA and ligands

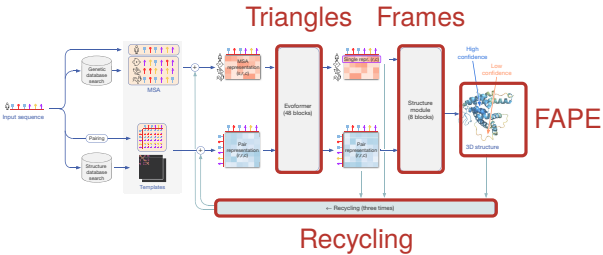
address the limits

Baker, Hassabis and Jumper receive the Nobel Prize in Chemistry

AlphaFold 3 is more accurate than specialised tools, with one exception

The first authors are Baek 2021, Tunyasuvunakool 2021, Evans 2021, Lin 2022 and Abramson 2024.

Summary



Evolution and geometry are trained end to end

One model of 384 residues takes about one GPU minute

The median backbone error is below one atom width

J. Jumper et al., “Highly accurate protein structure prediction with AlphaFold,” *Nature* 596, 583–589 (2021).

Exam question

Placeholder. The question from the course team arrives about seven days before the talk.

Our answer.

Questions for the audience

Placeholder. At least two questions, then a moderated discussion.

1. First question.
2. Second question.

Backup: the whole network as residual updates

	Update	Message	SI
Evoformer, 48 blocks			
1	$\mathbf{m} \leftarrow \mathbf{m} + \text{RowAttention}(\mathbf{m} \mid \mathbf{z})$	node → node, edge biases the weight	Alg. 7
2	$\mathbf{m} \leftarrow \mathbf{m} + \text{ColumnAttention}(\mathbf{m})$	across the sequences at one node	Alg. 8
3	$\mathbf{m} \leftarrow \mathbf{m} + \text{MLP}(\mathbf{m})$	per node	Alg. 9
4	$\mathbf{z} \leftarrow \mathbf{z} + \text{OuterProductMean}(\mathbf{m})$	node pair → edge	Alg. 10
5	$\mathbf{z} \leftarrow \mathbf{z} + \text{TriangleMultiplication}(\mathbf{z})$, out and in	edge pair → edge	Alg. 11, 12
6	$\mathbf{z} \leftarrow \mathbf{z} + \text{TriangleAttention}(\mathbf{z})$, start and end node	edge pair → edge	Alg. 13, 14
7	$\mathbf{z} \leftarrow \mathbf{z} + \text{MLP}(\mathbf{z})$	per edge	Alg. 15
Structure module, 8 blocks with shared weights. At the start $\mathbf{s}_u = \mathbf{m}_{1u}$ and T_u is the identity.			
8	$\mathbf{s} \leftarrow \mathbf{s} + \text{IPA}(\mathbf{s}, \mathbf{z}, T)$	node → node, values moved by $T_u^{-1} \circ T_v$	Alg. 22
9	$\mathbf{s} \leftarrow \mathbf{s} + \text{MLP}(\mathbf{s})$	per node	Alg. 20
10	$T_u \leftarrow T_u \circ \text{BackboneUpdate}(\mathbf{s}_u)$	per node, in its own frame	Alg. 23
Recycling, 4 cycles			
11	$\mathbf{z} \leftarrow \mathbf{z} + \text{Embed}(d_{uv})$, C$\beta$ distances from the frames	frames → edges	Alg. 32

Every line is $\mathbf{h}_u = \phi(\mathbf{x}_u, \bigoplus_v \psi)$ on nodes or on edges. Information moves in lines 1, 4, 5, 6, 8 and 11.